

# Statistics II: Introduction to Inference

## Problem set 1

### [General Problems]

1. Let  $X \sim \text{binomial}(n, p)$ . Find the p.m.f. of  $T = 2X + 1$ .
2. Let  $X \sim \text{normal}(0, 1)$ . Find the distribution of  $T = \text{sign}(X) = \mathbb{I}(X > 0)$ .

### [Distribution Function Technique]

1. Let  $X$  be distributed as some absolutely continuous distribution with cdf  $G_X$ , then show that  $T = G_X(X) \sim \text{uniform}(0, 1)$ .
2. Let  $X \sim \text{Cauchy}(0, 1)$  distribution, then show that  $T = 1/(1 + X^2) \sim \text{beta}(0.5, 0.5)$ .
3. Let  $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{uniform}(0, 1)$ , then find the distributions of  $X_{(1)}$  and  $X_{(n)}$ .
4. Let  $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{exponential}(\lambda)$ , then find the distributions of  $X_{(1)}$  and  $X_{(n)}$ .
5. Let  $X \sim \text{normal}(0, 1)$ . Find the p.d.f. of  $T = |X|$ .  
Let  $X \sim \text{uniform}(-1, 1)$ . Find the p.d.f. of  $T = X^2 + 3$ .

### [MGF Technique]

1. (**Additive properties**) *Prove the following statements using moment generating function.*
  - (a) Let  $X_i \stackrel{ind}{\sim} \text{binomial}(n_i, p)$ , for  $i = 1, \dots, k$ , then  $T = \sum_{i=1}^k X_i$  follows  $\text{binomial}(\sum_i n_i, p)$ .
  - (b) Let  $X_i \stackrel{ind}{\sim} \text{Poisson}(\lambda_i)$ , for  $i = 1, \dots, n$ , then  $T = \sum_{i=1}^n X_i$  follows  $\text{Poisson}(\sum_i \lambda_i)$ .
  - (c) Let  $X_i \stackrel{ind}{\sim} \text{normal}(\mu_i, \sigma_i^2)$ , for  $i = 1, \dots, n$ , then  $T = \sum_{i=1}^n X_i$  follows  $\text{normal}(\sum_i \mu_i, \sum_i \sigma_i^2)$ .
  - (d) Let  $X_i \stackrel{ind}{\sim} \text{Gamma}(\alpha_i, \beta)$ , for  $i = 1, \dots, n$ , then  $T = \sum_{i=1}^n X_i$  follows  $\text{Gamma}(\sum_i \alpha_i, \beta)$ .
  - (e) Let  $X_i \stackrel{ind}{\sim} \chi_{n_i}^2$ , for  $i = 1, \dots, k$ , then  $T = \sum_{i=1}^k X_i$  follows  $\chi_N^2$  where  $N = \sum_i n_i$ .

### [Transformation Technique]

1. Let  $X \sim \text{uniform}(0, 1)$  distribution, then  $T = -2 \log X \sim \chi_2^2$ .
2. Let  $X \sim \text{uniform}(0, 1)$  distribution, and  $\alpha > 0$  then  $T = X^{1/\alpha} \sim \text{beta}(\alpha, 1)$ .
3. Let  $X \sim \text{beta}(n/2, m/2)$  distribution, then show that  $T = mX/\{n(1 - X)\} \sim F_{n,m}$ .
4. Let  $X \sim \text{Gamma}(\alpha, \beta)$  distribution, then show that  $T = aX \sim \text{Gamma}(\alpha, \beta/a)$ .
5. Let  $X \sim \text{normal}(\mu, \sigma^2)$  distribution, then show that  $T = aX + b \sim \text{normal}(a\mu + b, a^2\sigma^2)$ .
6. Let  $X \sim \text{uniform}(0, 1)$ . Find the p.d.f. of  $T = X^2 + 3$ .